Spencer Daugherty

spencer.page.daugherty@gmail.com. spdaugherty.github.io. they/them.

Experience

Burnett Meyer Postdoctoral Fellow ..... 2024 -

University of Colorado Boulder. Mentor: Nat Thiem.

Education

Ph.D. Mathematics 2024

North Carolina State University. Advisors: Laura Colmenarejo, Sarah Mason.

Thesis: Schur-like bases and their colored generalizations.

Winston-Rose Award for Research. Maltbie Award for Teaching. Graduate Student Association Teaching Award.

B.A. Mathematics 2019

Mount Holyoke College. Minor Computer Science. Cum Laude.

Research Interests

Algebraic and Enumerative Combinatorics. Symmetric functions and their generalizations, combinatorial Hopf algebras, tableaux combinatorics, Catalan combinatorics, posets and lattices, combinatorial statistics.

Publications and Preprints

Enumeration of Latin tableaux and combinatorics of wide partitions. 2026

S. Daugherty, M. Fahnestock**, S. Huxtable**, M. Larson**, N. Al Munayes**, S. Peacock*. arXiv:2610.29291

The stable Tamari lattice 2026

H. Chau, S. Daugherty, S. Grate, M. Haiman, J. C. Martínez Mori, A. Pun.

Shuffle-compatibility for combinatorial statistics on words, parking functions, and set partitions. 2026

S. Daugherty, J. Liang. arXiv:2607.14255.

Lexical tableaux and quasisymmetric functions 2025

J. M. Campbell, S. Daugherty. arXiv:2511.00713.

Structure and geometry of the tableaux algebra 2025

S. Daugherty, N. González, B. Muniz*, P. Ocal, J. Pan, J. Torres. arXiv:2510.27209.

Dimensions of toggleability spaces 2025

B. Adenbaum, S. Daugherty, N. Mayers. arXiv:2508.14974. FPSAC 2026. (Revising for resubmission at Comb. Theory)

Metered parking functions 2025

S. Daugherty, P. Harris, I. Klein*, M. McClinton*. arXiv:2406.12941. INTEGERS.

A Hopf algebra generalization of the symmetric functions in partially commutative variables 2024

S. Daugherty*. arXiv:2412.11013.

Extended Schur functions and bases related by involutions 2024

S. Daugherty*. arXiv:2401.02502. Advances in Applied Mathematics. FPSAC 2024.

Combinatorics of generalized parking function polytopes 2025

M. Bayer, S. Borgwardt, T. Chambers*, S. Daugherty*, A. Dawkins*, D. Deligeorgaki*, H. Liao*, T. McAllister, A. Morrison*, G. Nelson*, A. Vindas-Meléndez. arXiv:2403.07387. Discrete & Computational Geometry.

A generalization of the dual immaculate functions in partially commutative variables 2024

S. Daugherty*. arXiv:2309.08518. Electronic Journal of Combinatorics.

Nimber sequences of Node-Kayles games 2020

S. Brown**, S. Daugherty**, E. Fiorini, B. Maldonado**, S. Rainville**, R. Waechter**, T. Wong. J. Integer Seq.

*graduate student at time of writing. **undergraduate student at time of writing.

Current Projects

Matrix shuffles and two-parameter quasisymmetric functions. .....

Combinatorial statistics and the q,t-Catalan numbers ..... with J. Lentfer.

Permutations in the stable Tamari lattice. ..... with H. Chau, S. Grate, J. C. Martínez Mori, A. Pun.

Teaching

Instructor of Record CU Boulder 2024-2026

Graduate topics in algebra. Discrete math. Linear algebra for non-math majors. 8 sections, 10-40 students per class.

Instructor of Record NCSU 2020-2024

Applied Differential Equations, Finite Math, Calculus II, Calculus III. 6 sections, 60-180 students per class.

Preparing the Professoriate Fellow NCSU 2023-2024

Professional development that gives doctoral students mentoring, teaching, and future faculty preparation experience.

Math Instructor NC Governor's School East 2022

Graph Theory. Four-week long course for gifted high school students.

Teaching Assistant NCSU 2019-2020

Calculus I, Calculus II, Applied Differential Equations. 3 semesters.

Talks and Posters

Upcoming Talks

Kennesaw State University Discrete Math Seminar talk ..... Nov 2026

AMS Western Sectional Special Session in Algebraic and Geometric Combinatorics talk ..... Nov 2026

JMM Special Session on Symmetric Functions, Hopf Algebras, and Combinatorics talk ..... Jan 2027

The stable Tamari lattice

University of Denver Algebra and Logic Seminar talk ..... Oct 23, 2026

CU Boulder Algebraic Lie Theory Seminar talk ..... Oct 20, 2026

AMS Western Sectional GEMS of Combinatorics Special Session talk ..... Oct 27, 2025

Shuffle-compatibility for words, parking functions, and set partitions

Great Plains Combinatorics Conference poster ..... May 2, 2026

Oregon State University Algebra and Number Theory Seminar talk ..... Apr 21, 2026

UC Berkeley Combinatorics Seminar talk ..... Mar 18, 2026

CU Boulder Algebraic Lie Theory Seminar talk ..... Mar 10, 2026

IAS Collaborative Workshop in Algebraic Combinatorics lightning talk ..... Jun 2, 2025

Dimensions of toggleability spaces

University of South Carolina Combinatorics Seminar talk ..... Sep 12, 2025

Metered parking functions

JMM Spectra Special Session talk ..... Jan 10, 2025

AMS Sectional Riverside: Special Session on Gender Equity in Combinatorics talk ..... Oct 27, 2024

Mount Holyoke College talk ..... Sep 26, 2024

Combinatorics of generalized parking function polytopes

JMM AMS Special Session on work from the Graduate Research Workshop in Combinatorics talk ..... Jan 8, 2025

NCSU Pure Graduate Student Seminar talk ..... Apr 15, 2024

University of North Carolina Wilmington Seminar talk ..... Mar 15, 2024

Combinatorial statistics and q,t-Catalan numbers

University of Washington Combinatorics and Geometry Seminar talk ..... Nov 20, 2024

NCSU Pure Graduate Student Seminar talk ..... Sep 18, 2023

Colored Schur-like bases and generalizations of the symmetric functions

CU Boulder Algebraic Lie Theory Seminar talk ..... Sep 17, 2024

Michigan State University Virtual Seminar talk ..... Apr 3, 2024

Extended Schur functions and bases related by involutions

Formal Power Series and Algebraic Combinatorics Bochum poster ..... Jul 25, 2024

Mid-Atlantic Algebra, Geometry, and Combinatorics Workshop poster ..... Dec 2, 2023

University of Waterloo Seminar talk ..... Nov 9, 2023

A partially commutative generalization of the dual immaculate functions

Okinawa Institute of Science and Technology talk ..... Feb 9, 2024

UAlbany Women in Algebra and Combinatorics poster ..... Apr 29, 2023

AlGeCom XIII poster ..... Apr 15, 2023

NCSU Pure Graduate Student Seminar talk ..... Nov 21, 2022

Schur-like Bases of NSym and QSym: Properties of the Shin Functions

Wake Forest Colloquium talk ..... Oct 19, 2023

NCSU Algebra and Combinatorics Seminar talk ..... Oct 18, 2023

CAGE Seminar talk ..... Sep 19, 2023

Chromatic symmetric functions of trees

Triangle Area Graduate Math Conference talk ..... Sep 30, 2023

Organizing and Service

Grant Coordinator for The Combinatorics Consortium 2025-

Organizer of CU Boulder Algebraic Lie Theory Seminar Boulder, Colorado 2024-

Organizer of JMM Special Session for the MRC in Algebraic Combinatorics Seattle, Washington ... Jan 9, 2025

Reviewer: Discrete Mathematics & Theoretical Computer Science, Electronic Journal of Combinatorics, American Journal of Combinatorics, Discrete Mathematics, Advances in Applied Mathematics.

Workshop and Program Participation

Collaborate@ICERM ..... July 2025

IAS Summer Workshop in Algebraic Combinatorics ..... June 2025

MRC in Algebraic Combinatorics Java, New York ..... June 2024

AlCoVR Workshop in Dynamical Algebraic Combinatorics Virtual Reality ..... May 2024

Graduate Research Workshop in Combinatorics Laramie, Wyoming ..... July 2023